S2 Table. PLS guidelines and criteria.

Guideline	Linguistic Attributes	Formal Attributes	General Content	Presentation of Results	Presentation of Quality of Evidence	Contextual Information
American Psychological Association ("translational abstract"), psychological science	Tone: ▪ more personal and friendly than scientific abstract ▪ do not overstate / oversimplify findings or conclusions Words: - Sentences: -	Text Length: ▪ 150 - 200 words Text Structure: - Use of Tables/Visuals: -	Title: - Content structure: ▪ follow a pattern of introduction, method, results, discussion Headlines: - Content: ▪ consider your audience ▪ clearly describe investigated problem ▪ describe participants only insofar as relevant to the audience; details of the sample only if it is remarkable in some way ▪ detailed information about study methods may be summarized or omitted ▪ emphasize conclusions that are relevant for the audience ▪ try to create a "take home message"	▪ state the findings in clear, nontechnical language ▪ remove any statistics ▪ if article contains more multiple studies, summarize contents of all studies	-	Purpose of text: ▪ clear communication of article content ▪ emphasis on article's value to educated public/professional audiences Author of PLS: - Review of PLS: - Access: - Context: ▪ recommendation for PLS as standard feature of each APA journal article
Cochrane 2013, standards for the reporting of PLS ("Plain Language Summaries"), Cochrane Intervention Reviews	Tone: - Words: ▪ avoid technical terms and jargon (explain them if unavoidable) ▪ avoid long words / words with many syllables / misunderstandable words ▪ consider introducing an acronym or short term for repeated use ▪ avoid regional terms	Text Length: ▪ 250 - 300 words Text Structure: ▪ use short paragraphs Use of Tables/Visuals: -	Title: ▪ restate review title in plain language if possible, otherwise explain terms ▪ avoid recommendations Content structure: ▪ review question ▪ background ▪ study characteristics ▪ key results ▪ quality of evidence Headlines:	▪ present results for all main outcomes ▪ use consistent wording across outcomes ▪ report findings on harms that are described in the review and state whether they have been fully reported in the included studies ▪ it is not essential to provide numerical data ▪ do not present	▪ describe overall quality of evidence for each of the main outcomes based on GRADE considerations ▪ describe any factor that could affect the confidence in the results ▪ provide key reasons for quality of evidence / limitations in lay terms ▪ describe if quality of evidence is high	Purpose of text: ▪ summary of the review that contains the crucial information in plain language and that will be understood by the general public Author of PLS: - Review of PLS: - Access: - Context: ▪ tailor messages across different summary versions of the review

	(AE / BE) ▪ use active voice Sentences: ▪ one keypoint / sentence ▪ avoid more than two hard words in a sentence unless you can explain them		▪ standard headings ▪ consistent order ▪ in bold type Content: ▪ convey the question addressed in the review ▪ shortly describe population, intervention and outcomes in Background section ▪ give enough detail on study characteristics (incl. search date and population details)	numerical data if estimations of effects are imprecise or uncertain ▪ if numerical data is provided, use natural frequencies for dichotomous outcomes and accompany relative effects with absolute effect estimates ▪ explain any statistical term ▪ statics, if used, should provide valid, digestible summary of direction, size and precision of the effect estimates	▪ if impact of funding sources on quality of the evidence is considered in the review, include a statement in the PLS	▪ consistent reporting of key messages between PLS, main text, SoF table and authors' conclusions.
Cochrane 2016, Checklist for PLS Review ("Plain Language Summaries"), Cochrane Intervention Reviews	Tone: - Words: ▪ avoid or explain acronyms, jargon and technical terms ▪ PLS authors can choose to refer to "we" or to "review authors", but be consistent Sentences: -	Text Length: ▪ 400-700 words Text Structure: - Use of Tables/Visuals: -	Title: ▪ if title is difficult to understand, consider re-writing it in plain language Content structure: ▪ What is the aim of review? ▪ Key messages ▪ What was studied in the review? ▪ What are the main results of the review? ▪ How up-to-date is this review? Headlines: ▪ use sub-headings Content: ▪ explain that results come from a systematic study rather than a single study ▪ describe if necessary: why this topic is important; the population; the	▪ present results only for the most important outcomes, try not to present more than 7 outcomes ▪ if no data was found, present outcomes nevertheless ▪ present results consistently (similar words / expressions for similar effects) ▪ use the provided standard sentences ▪ use absolute numbers (not relative risks, odds ratios, or percentages)	▪ present quality or certainty for each outcome as presented in the SoF table ▪ if quality/certainty is not high, avoid strong statements, add modifying statements	Purpose of text: - Author of PLS: - Review of PLS: ▪ the checklist is intended for use by reviewers of PLS Access: - Context: ▪ a template, PLS examples and standardized statements are provided ▪ ensure results are reported consistently between PLS and main text / abstract / SoF table

			intervention; the comparison group; the outcomes; possible adverse effects ▪ do not present recommendations			
Cochrane 2019 [1] ("Plain Language Summaries"), Cochrane Intervention Reviews	Tone: - Words: ▪ avoid research jargon ▪ refer to "study" rather than "trial" ▪ use name of the outcome and name of intervention instead of "outcome" / "intervention" ▪ use words like "parents", "women", etc. instead of "participants" Sentences: -	Text Length: ▪ *see above* Text Structure: - Use of Tables/Visuals: ▪ if PLS is published outside Cochrane Library, add a simplified version of SoF table	Title: ▪ *see above* Content structure: ▪ *see above* Headlines: ▪ *see above* Content: ▪ describe if necessary: why this topic is important; the population / health problem addressed in the review; the intervention and what it was compared to; the outcomes. ▪ explain that results come from a systematic study rather than a single study ▪ do not present recommendations	▪ *see above* ▪ use standardized qualitative statements when reporting effects of an intervention ▪ ideally, use numbers and words; if you use numbers, present them in parentheses after the qualitative statement ▪ use absolute numbers (not relative risks, odds ratios, percentages or numbers needed to treat) ▪ when presenting continuous outcomes using numbers, refer to the scale	▪ *see above* ▪ presenting confidence intervals is mostly not necessary but it may be useful in some situations (example statement is provided)	Purpose of text: - Author of PLS: - Review of PLS: - Access: - Context: ▪ *see above*
The Steering Group of the Campbell Collaboration (2016) ("Plain Language Summaries"), Campbell systematic reviews	Tone: ▪ use direct language ▪ accessible manner Words: ▪ avoid jargon ▪ specific terms that are familiar to policy makers and practitioners should be retained ▪ verbs: present tense Sentences: -	Text Length: ▪ 600 - 750 words Text Structure: - Use of Tables/Visuals: ▪ text box: "What is the aim of this review?" Form: ▪ no footnotes or references	Title: ▪ headline style ▪ summarizing main findings ▪ reference to full title at the end of the PLS Content structure: ▪ see "Headlines" Headlines: ▪ use headings and additional subheadings if needed ▪ Headlines: The Review in Brief; What is this Review about?; What are the main findings of this	▪ report the study findings directly and in present tense ▪ avoid selective outcome reporting ▪ use qualitative statements when presenting the effects of the intervention ▪ avoid numbers to increase accessibility ▪ qualitative statements: similar words and expressions for similar levels of effect (additional appendix with standardised	▪ included studies: optionally add statement about the quality of evidence ▪ results: GRADE reporting system ▪ decide whether the size of the effect is important, less important or not important to the user ▪ use standardised statements about effect ▪ presenting confidence intervals is mostly not necessary but it may be useful in some	Purpose of text: ▪ to make information about main findings available in an easily understandable format ▪ to report this information in a consistent way Author of PLS: ▪ study authors submit PLS with final review Review of PLS: ▪ edited by the secretariat, revised version checked with lead study author ▪ meta-information

			review?; What do the findings of this review mean?; How up-to-date is this review?; What is the Campbell Collaboration?; About this summary Content: ▪ assessed interventions (i.a.) ▪ primary outcomes ▪ results for each outcome and analysis of heterogeneity ▪ discussion of theory of change (i.a.) ▪ quality of evidence ▪ implications for policy, practice, research (i.a.)	statements)	situations (standardized statement is provided)	reported as last sentence in PLS Access: - Context: ▪ results should be reported consistently between PLS and main text / abstract ▪ a template with suggested wording is provided
Expert group on clinical trials for the implementation of Regulation (EU) No 536/2014 ("lay summary" / "summary of clinical trial results for laypersons"), clinical trial results	Tone: ▪ avoid any promotional language Words: ▪ avoid jargon / technical / medical / scientific language ▪ remove unnecessary / complex words ▪ be consistent with use of words and define them ▪ explain underlying concepts if necessary ▪ avoid ambiguous words / phrases ▪ limit use of acronyms, and of abstract or multisyllabic words ▪ verbs: use active rather than passive voice ▪ texts should be aimed at literacy proficiency level of 2-3 (use metrics to measure text readability) Sentences:	Text Length: ▪ as short as possible Text structure: ▪ bullet points ▪ white space between topics ▪ minimum of 12-point font ▪ avoid text in all caps or underlining Use of Tables/Visuals: ▪ use visuals to convey critical concepts but limited use of unnecessary images ▪ visuals should present one message per image and be clearly labelled with captions	Title: ▪ specific to the trial ▪ may be shortened / simplified Content structure: ▪ present general information before the details Headlines: ▪ use headlines and descriptive subheadings to organize information Content: ▪ focus on unambiguous, factual information ▪ no promotional content should be included ▪ avoid overwhelming with too much information ▪ describe adverse reactions	▪ describe every study arm and outcome ▪ Numbers: follow numeracy principle (links to additional information is provided) ▪ report frequencies both in numerical terms and percentages ▪ report all outcomes	▪ don't include promotional content	Purpose of text: ▪ EU Clinical Trials Regulation 536/2014 (Article 37): sponsors need to provide summary results in a format understandable to laypersons (research participants and general public) Author of PLS: - Review of PLS: ▪ consider including patients, patient representative or advocates to ensure the summary meets its aims; consider help of medical writers with plain language expertise Access: ▪ made available in the EU Portal and Database Context: ▪ provide summary as a

	▪ make short and succinct sentences instead of long and complex ones					minimum in local language of where the trial took place ▪ where possible include English version ▪ a template with 10 elements that should be included in the summary and suggested wording is provided
Lionbridge (2019): Seven rules for effective communication ("Plain Language Summaries"), clinical trial results	Tone: ▪ information should be presented factually and neutrally ▪ respectful tone so that study participants do not feel as victims; avoid marketing jargon or overly positive presentation Words: ▪ simple and unambiguous ▪ avoid polysyllabic words; complex technical terms (unless you explain them); scientific jargon that may be misunderstood or cause confusion Sentences: ▪ active sentences with a subject that executes an action ▪ short, simple sentences ▪ avoid subclauses and nested sentences	Text Length: - Form: - Use of Tables/Visuals: ▪ use visuals aids such as empty spaces or graphs if it enhances communication with the target group	Title: - Content structure: - Headlines: - Content: ▪ content and presentation should be oriented towards language, style and knowledge of laypersons	▪ report absolute, whole numbers ▪ report numbers without decimal places or as percentages ▪ avoid probabilities or relative risks	-	Purpose of text: ▪ to inform persons with average education level about results and conclusions of clinical studies Author of PLS: - Review of PLS: - Access: ▪ access to extensive guidelines for writing PLS only for members Context: -
TransCelerate Biopharma Inc. ("lay summary"), clinical trial results	Tone: ▪ factual and objective tone Words: ▪ avoid superlative and enthusiastic words ▪ use neutral language	Text Length: - Form: ▪ use material that is fair and balanced in terms of formatting Use of Tables/Visuals: ▪ do not use materials	Title: - Content structure: - Headlines: - Content: ▪ factual and objective content ▪ use accurate,	▪ make only comments on the outcome that are factual in nature ▪ do not make inferences or assessments ▪ link to additional neutral language guide	▪ include a statement that results are from a single trial and different results may be obtained from other studies ▪ include a statement that	Purpose of text: ▪ written to be understandable to the general public ▪ recommendations to help prepare PLS that are not perceived as

	(link to additional guidance is provided) Sentences: -	that have a commercial appearance ▪ do not use brand colors / logos ▪ be fair and balanced in terms of formatting	non-misleading information ▪ include information on efficacy and safety data from the trial ▪ do not provide approval status ▪ material should be fair and balanced	is provided	no therapeutic changes should be made based on the results of a single trial without consulting a healthcare professional	promotional Author of PLS: ▪ sponsor of trial Review of PLS: - Access: ▪ posted to public websites, EU database ▪ provided to trial participants Context: ▪ take care that PLS is only provided in a non-promotional context ▪ add a statement with link to ClinicalTrials.gov & EU clinical trial register
Duke (2012): How to write a lay summary ("lay summary"), Digital Curation Center collaborating with Patients Participate! project	Tone: ▪ do not write to entertain Words: ▪ use everyday English words instead of complex words ▪ avoid meaningless terms ▪ use active voice and second person instead of third person ▪ use person-centered language, do not focus on circumstances, illness or disability ▪ write in an easily readable style Sentences: ▪ use short and clear sentences ▪ avoid complex grammatical structures ▪ phrase positively rather than negatively	Text Length: ▪ adhere to convention Form: ▪ break text up into paragraphs Use of Tables/Visuals: -	Title: ▪ provide a good and relevant title Content structure: ▪ order the text logically and let it flow naturally ▪ adhere to convention ▪ provide a first sentence that gives a concise introduction into text Headlines: - Content: ▪ provide answers to the questions: Who, What, Where, When, Why, How? ▪ give concrete everyday examples ▪ make sure risks are appropriately communicated (i.a.)	-	-	Purpose of text: ▪ to shortly report research to a general audience Author of PLS: ▪ researchers Review of PLS: - Access: ▪ service providers may be responsible for the dissemination of research findings Context: -

Dubé & Lapane (2014): Lay Abstracts and Summaries: Writing Advice for Scientists ("lay abstract" / "summaries"), multidisciplinary	Tone: - Words: ▪ avoid long / multisyllabic / complicated words ▪ do not use acronyms (except commonly known ones) ▪ use active instead of passive voice ▪ check readability and reading level Sentences: ▪ shorten sentences (but avoid choppy writing)	Text Length: ▪ depends on purpose Form: ▪ check instructions Use of Tables/Visuals: -	Title: ▪ state the main impact of your work for your audience with a simple phrase Content structure: ▪ organize to make your story clear ▪ summarize the most important / relevant information at the beginning briefly Headlines: - Content: -	-	-	Purpose of text: ▪ the goal: a summary that is accessible to the public while true to science Author of PLS: - Review of PLS: ▪ review from target audience and other scientists of your field Access: - Context: -
eLife (2017): Plain-language summaries: How to write an eLife digest ("Plain Language Summaries" / "eLife digests"), life sciences	Tone: - Words: ▪ use words that are understandable to the widest group of readers ▪ avoid technical jargon (note that jargon can include common words that are used in a field-specific way ▪ if scientific terms must be used, define each at first use in more everyday-language ▪ only use well-known acronyms and do not use more than 3 acronyms overall ▪ rather use a few short words than one long one ▪ use verbs instead as nouns as much as possible	Text Length: - Form: - Use of Tables/Visuals: -	Title: - Content structure ▪ First sentence: Include something that most readers will be able to relate to ▪ Following sentences: Get gradually more specific Headlines: - Content: ▪ 4 questions that should be answered: 1. What background information would someone who is completely unfamiliar with your field need to know to understand the findings in your paper? 2. What exact research question did you set out to answer and why?	▪ focus on findings highlighted in the title / abstract of your paper; explain them clearly and completely ▪ describe methodology with 1-2 sentences ▪ always mention which species, type of organism or cells you have studied	▪ avoid hype / exaggeration	Purpose of text: ▪ explain findings of a eLife paper to a broader audience Author of PLS: ▪ written by editors and writers working together with authors: the author of the scientific paper is asked to answer 4 questions; from that, the eLife Features team writes the digest Review of PLS: - Access: - Context: ▪ published in a prominent position ▪ some are republished on social media platform ▪ encouragement to re-use PLS in other places

	Sentences: ▪ active sentences ▪ all sentences should be shorter than 35 words		3. What are the most important findings of your paper? (always mention which species, type of organism or cells you have studied) 4. Who might eventually benefit from the findings of your study, and what would need to be done before we could achieve these benefits?			
NIHR ("Plain English Summaries"), health research	Tone: ▪ clearly and simply written Words: ▪ avoid jargon, abbreviations and technical terms (provide explanation if you must use them) ▪ active rather than passive phrases ▪ link to glossary with lay-friendly terms Sentences: ▪ keep short	Text Length: ▪ depends on funding programme Form: ▪ break up the text, e.g. with bullet lists or headings Use of Tables/Visuals: -	Title: - Content structure: ▪ include, where appropriate, information on: Aim(s) of the research; Background of research; Design and methods used; Patient and public involvement; Dissemination of the findings Headlines: - Content: -	-	-	Purpose of text: - Author of PLS: ▪ strong advice to involve patients, carers and/or members of the public Review of PLS: - Access: - Context: ▪ links to resources are provided ▪ summary will be used on NIHR and other websites to describe the research
American Geophysical Union [several journals] ("Plain Language Summaries"), earth and space science	Tone: ▪ language and tone are different from scientific abstract Words: ▪ avoid jargon, incl. undefined / excessive acronyms, field-specific terms, obscure / unnecessary long words, words that have different meaning to non-scientists ▪ use straight-forward descriptions Sentences: -	Text Length: - Form: - Use of Tables/Visuals: -	Title: - Content structure: - Headlines: - Content: ▪ develop your take home message by explaining what the research is about, what you found, and why it matters/what the impact is. ▪ provide enough context for those outside of your specific area of science will need more context ▪ highlight the novelty	-	-	Purpose of text: ▪ to explain your science to broader audiences Author of PLS: ▪ authors of the scientific article Review of PLS: - Access: - Context: -

			and value of your research ▪ Think about: What was the research question (in the larger context of your field)? What did your study find? Why does it matter? What's the take-home message?			
Halprin (2021)[1]: Plain Language Summaries Explained in Plain Language, [several journals] ("Plain Language Summaries"), earth and space science	Tone: - Words: ▪ *see above* ▪ define any field-specific terms ▪ explain acronyms if they have to be used Sentences: -	Text Length: ▪ 200 words Form: - Use of Tables/Visuals: -	Title: - Content structure: - Headlines: - Content: ▪ recommendation to use four elements (use 1-2 sentences each): 1) Topic Overview; 2) Paper Overview; 3) Findings Summary; 4) Key Takeaways	-	-	Purpose of text: ▪ *see above* Author of PLS: ▪ *see above* Review of PLS: - Access: - Context: ▪ examples are provided
People and Nature [journal] ("Plain Language Summaries"), human-ecology interactions	Tone: ▪ use clear and simple language ▪ simple and straightforward style Words: ▪ avoid jargon ▪ avoid scientific terms; if you must use a scientific term, explain in clear and simple terms what it is Sentences: -	Text Length: ▪ 250-300 words Form: - Use of Tables/Visuals: ▪ try and include a photo or image, anything that makes you work more accessible and interesting	Title: ▪ plain language title that says what the paper is about in clear terms (a bit like a newspaper headline) ▪ short (< 120 characters) Content structure: - Headlines: - Content: ▪ include any important contextual background or findings that might make your work more relevant, interesting or memorable to the reader	-	-	Purpose of text: ▪ to raise the profile of your work ▪ to make it accessible to the widest possible audience Author of PLS: ▪ author of paper Review of PLS: ▪ upload PLS at revision stage with your work ▪ will be edited and returned to author if too much jargon Access: ▪ freely available to read Context: ▪ PLS will be added to journal blog, included with the paper ▪ encouraged to reuse PLS by posting it on own blog, share on social media

Autism [journal] ("lay abstract"), all aspects of autism spectrum disorders and related developmental disabilities	Tone: ▪ easily understandable Words: ▪ avoid technical terminology Sentences: -	Text Length: ▪ max. 250 words Form: - Use of Tables/Visuals: -	Title: - Content structure: - Headlines: - Content: ▪ Consider the questions: What is already known about the topic? What does this paper add? Implications for practice, research and policy	▪ avoid reporting of statistics	-	Purpose of text: - Author of PLS: ▪ author of paper Review of PLS: ▪ provide lay abstract as part of submission Access: ▪ will be made widely available to general public and particularly to autistic people and their families Context: -
Taylor & Francis Group [journal] ("Plain Language Summaries"), multidisciplinary	Tone: ▪ as objective as possible ▪ person-centered rather than focusing on the condition/illness Words: ▪ free of jargon ▪ clear language ▪ easy-to-understand language instead of complex words ▪ active rather than passive voice ▪ avoid grammatical structures and abbreviations ▪ if you need to use a technical term, explain it the first time it is used Sentences: ▪ short clear sentences ▪ phrase sentences in neutral language	Text Length: ▪ max. 250 words Form: ▪ continuous prose format Use of Tables/Visuals: -	Title: - Content structure: - Headlines: - Content: ▪ Answer the questions: Why was the study done? What did the researchers do and find? What do the results mean? What is the objective influence on the wider field?	▪ keep statements factual ▪ avoid providing opinions / speculation on study results and significance ▪ priority is that PLS is not misleading	-	Purpose of text: ▪ to communicate the significance of scientific research evidence to a broad evidence Author of PLS: ▪ author of scientific article submitted to journal Review of PLS: ▪ PLS is peer reviewed alongside the manuscript Access: ▪ Open access license is the same as that of the published article ▪ freely available to read on website Context: ▪ submit PLS as part of the manuscript or one year after (then incurring a fee) ▪ submission of PLS with scientific article is optional, but encouraged

i.a. = if applicable; GRADE = Grading of Recommendations Assessment, Development and Evaluation (specific formal process for grading systematic reviews); SoF = Summary of Finding table (standardized form to report results of Cochrane Reviews)

[1] listed are only changes to the version mentioned above